Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/13/2016
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Spaceflight Effects on Neurocognitive Performance: Extent, Longevity, and Neural Bases		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-294-1722
Organization Name:	University of Florida		
PI Address 1:	Applied Physiology & Kinesiology		
PI Address 2:	FLG 142, P.O. Box 118205		
PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	08/26/2011	**End Date:**	07/13/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 7/13/2017 due to transfer to PI's new institution (Ed., 1/11/2018) NOTE: Extended to 3/31/2019 per NSSC information and PI (Ed., 8/20/2012)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Bloomberg, Jacob (NASA Johnson Space Center) Mulavara, Ajitkumar (Universities Space Research Association) Wood, Scott (NASA Johnson Space Center)		
Grant/Contract No.:	NNX11AR02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Research Annoucement (NRA) NNJ10ZSA003N requested proposals to assess changes in elemental neurocognitive functions such as perception, motor control, memory, attention, language, executive function, and emotional processing following long duration spaceflight using both behavioral assessments and monitoring technologies such as functional magnetic resonance imaging (fMRI). In response to this call, we propose to perform structural and functional MR brain imaging to identify the relationship between changes in crewmember neurocognitive function and neural structural alterations following a six month International Space Station (ISS) mission. Our central hypothesis is that measures of brain structure, function, and network integrity will change from pre to post flight in crewmembers (Aim 1). Moreover, we predict that these changes will correlate with indices of cognitive, sensory, and motor function in a neuroanatomically selective fashion (Aim 2). Our interdisciplinary approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that will be conducted pre flight, during flight, and post flight to investigate neuroplastic and maladaptive brain changes in crewmembers following long duration spaceflight. Success in this endeavor would 1) result in identification of the underlying neural mechanisms and operational risks of spaceflight-induced changes in behavior, and 2) identify whether a return to normative behavioral function following re-adaptation to Earth's gravitational environment is associated with a restitution of brain structure and function or instead is supported by substitution with compensatory brain processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this project will have relevance not only to understanding the effects of spaceflight on the human brain and behavior, but also for delineating the capacity of the brain to remodel in response to adaptive stimuli. As such, the results should prove informative for understanding the neural mechanisms associated with adaptive behavioral change and the rehabilitation of these changes during recovery periods.
Task Progress:	Year 1. The project received partial funding during the first selection year "Definition Phase." During the first year "Definition Phase," the team was very productive. We attained Institutional Review Board approval from both the University of Michigan and NASA, which means that the project risks and benefits have been vetted through independent review boards. We also created and had our Experiment Document (ED) approved by the International Space Station Medical Projects (ISSMP) CCB. The ED is a technical report which outlines all of the methods we will utilize for this project. It is referred to throughout the course of the study by numerous groups as the "gold standard" for the experiments. The project has received "select for flight" on May 30, 2012. This means that we have undergone a feasibility review to confirm that the experiments can be implemented with crewmembers on the International Space Station. In this project, we will measure various aspects of motor, sensory, and cognitive function in crewmembers (and a group of control subjects) at multiple points prior to, during, and following spaceflight. Measures of brain structure and function will also be obtained pre and post flight. During the first project year, we have been purchasing, setting up, and debugging various hardware components with which to obtain these measurements. We have also been writing, refining, and testing our software for running these experiments. In the final months of the first funding year, we will run the initial pilot subjects through our behavioral and brain imaging protocols. This will enable us to ensure that we are obtaining good, clean data from which we can calculate our measures of interest prior to study initiation. Year 2. During the past year we continued to test and refine our measures. Our complementary, National Space Biomedical Research Institute (NSBRI) funded bed rest study is up and running with four subjects enrolled to date. We will utilize the same measures in the current study. This means that we have our data collection and analysis pipelines in place and ready to support flight data acquisition. Moreover, our trouble shooting has resulted in our selection of a head tapper device to stimulate the functional vestibular cortex in the MRI scanner, as opposed to our initial plan of using the auditory click method. The clicks need to be delivered at a high decibel which can be somewhat unpleasant for subjects. We have also obtained all of the necessary approvals for our inflight testing software, and it has been delivered for implementation with the next ISS software build. The PI (Principal Investigator) has obtained both WebMirage and Launchpad accounts for data transfer. In addition, one crewmember has given their informed consent to participate in this study for increment 39-40. This project also includes a ground based normative control study. We will initiate testing for this arm of the project within approximately the next month. Year 3. This past year was very successful with the completion of data collection for the bed rest (n=18) and ambulatory control (n=16) groups. We are in the process of analyzing both data sets and preparing manuscripts. The results will greatly enhance our interpretation of flight study data. Given this ability to leverage data collection from our ongoing NSBRI-funded bed rest study, we have extensively refined our data collection and analysis approaches. We have obtained all of our necessary ethical, hardware, and software approvals. Our first flight participant recently returned from the ISS and we have completed the first post flight data collection session. We have pre tested a few additional flight subjects and are anticipating smooth data collection and analysis processes given our experience with the bed rest and normative control subjects. Year 4. This past year has been one of steady data collection and analyses. We have collected or are in the process of collecting data on 7 crewmembers, including for a one-year mission participant. We have also obtained some retrospective MRI metrics from the Lifetime Surveillance of Astronaut Health (LSAH) (and are analyzing these data to provide greater insight for interpretation of our prospective study data.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Articles in Peer-reviewed Journals	Seidler RD, Mulavara AP, Bloomberg JJ, Peters BT. "Individual predictors of sensorimotor adaptability." Front Syst Neurosci. 2015 Jul 6;9:100. eCollection 2015. Review. http://dx.doi.org/10.3389/fnsys.2015.00100 ; PubMed PMID: 26217197; PubMed Central PMCID: PMC4491631 , Jul-2016
Articles in Peer-reviewed Journals	Koppelmans V, Mulavara AP, Yuan P, Cassady KE, Cooke KA, Wood SJ, Reuter-Lorenz PA, De Dios YE, Stepanyan V, Szecsy DL, Gadd NE, Kofman I, Scott JM, Downs ME, Bloomberg JJ, Ploutz-Snyder L, Seidler RD. "Exercise as potential countermeasure for the effects of 70 days of bed rest on cognitive and sensorimotor performance." Front Syst Neurosci. 2015 Sep 3;9:121.eCollection 2015. http://dx.doi.org/10.3389/fnsys.2015.00121 ; PubMed PMID: 26388746; PubMed Central PMCID: PMC4558429 , Sep-2015

Articles in Peer-reviewed Journals	Hirsiger S, Koppelmans V, Mérillat S, Liem F, Erdeniz B, Seidler RD, Jäncke L. "Structural and functional connectivity in healthy aging: Associations for cognition and motor behavior." Human Brain Mapping. 2016 Mar;37(3):855-67. Epub 2015 Dec 14. http://dx.doi.org/10.1002/hbm.23067 ; PubMed PMID: 26663386 , Mar-2016
Articles in Peer-reviewed Journals	Cassady K, Koppelmans V, Reuter-Lorenz P, De Dios Y, Gadd N, Wood S, Castenada RR, Kofman I, Bloomberg J, Mulavara A, Seidler R. "Effects of a spaceflight analog environment on brain connectivity and behavior." Neuroimage. 2016 Nov 1;141:18-30. Epub 2016 Jul 14. http://dx.doi.org/10.1016/j.neuroimage.2016.07.029 ; PubMed PMID: 27423254 , Nov-2016